

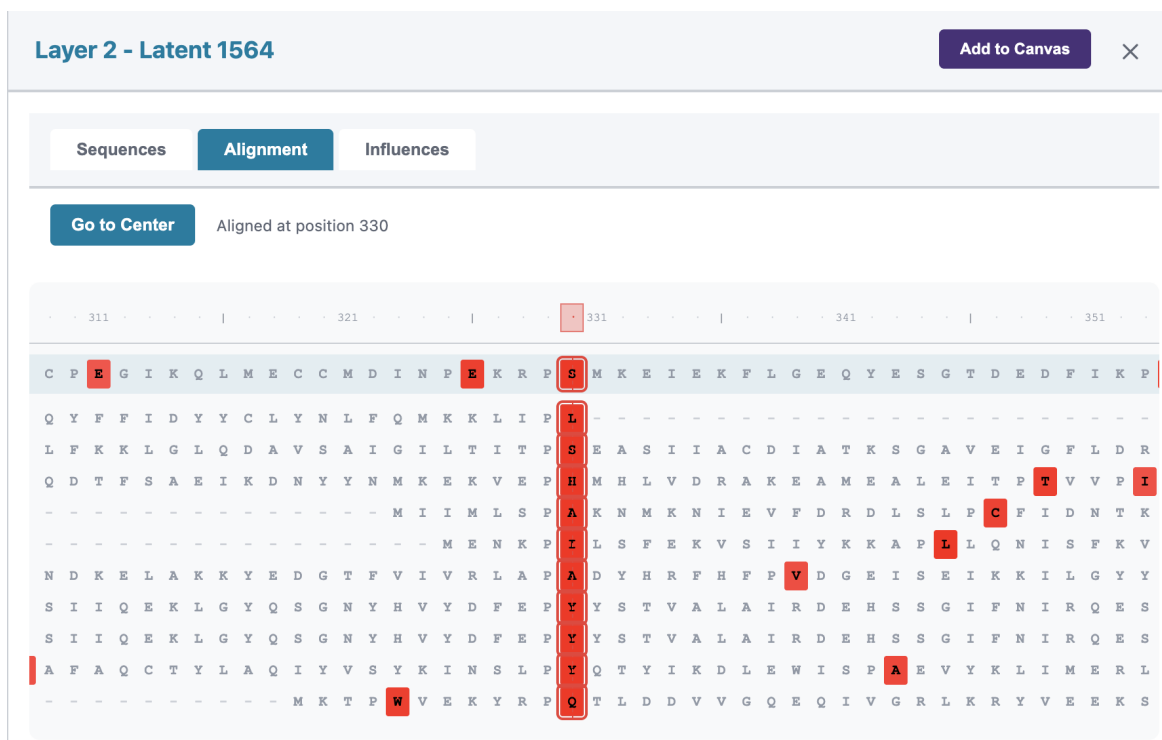


Figure 12. Latent Information Panel: Alignment tab. The Alignment tab displays an alignment between the input sequence (top row) and top-activating sequences from Swiss-Prot. Activation intensities are shown as colored overlays, with red indicating high activation. The “Go to Center” button navigates to the position of maximum activation of wild type and top activating sequences. This view facilitates identification of conserved sequence motifs and structural features recognized by the latent.

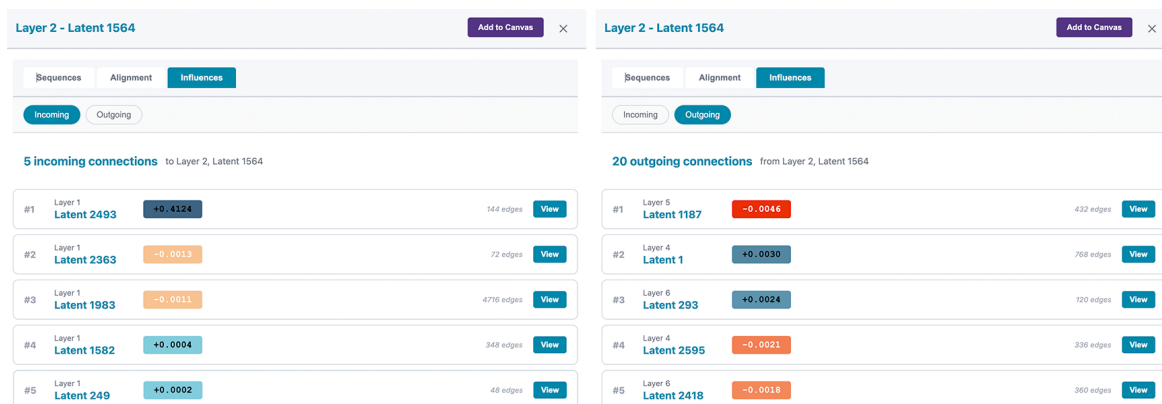


Figure 13. Latent Information Panel: Influences Tab. The Influences tab reveals the circuit connectivity of a selected latent. The left panel shows incoming connections from earlier layers, while the right panel shows outgoing connections to later layers. Each connection displays the source/target layer and latent index, the virtual weight (green for positive, orange for negative), and the number of sequence positions where this edge is active.

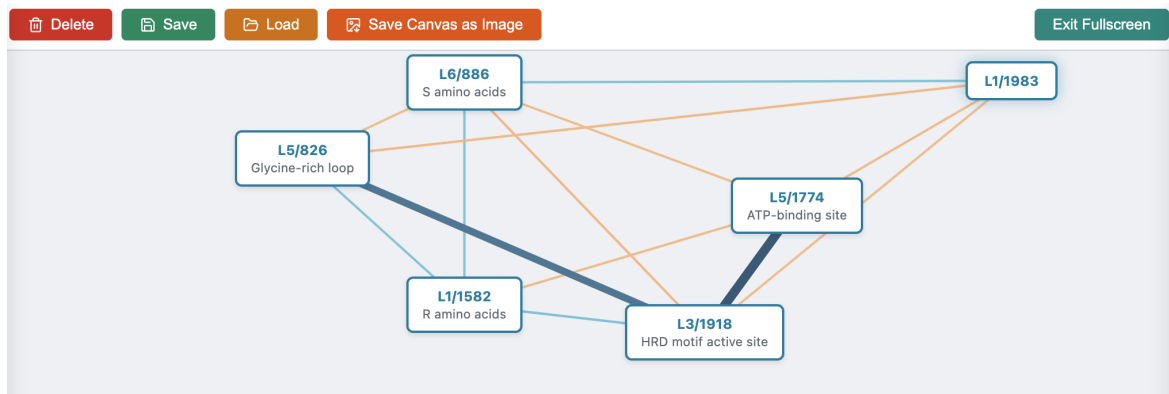


Figure 14. Interactive Circuit Canvas. The canvas provides a workspace for constructing and annotating circuit diagrams. Nodes represent latents (labeled as Layer/Index, e.g., “L5/1774”) and can be annotated with descriptive names documenting their hypothesized function (e.g., “ATP-binding site”, “Glycine-rich loop”). Adding a name to a latent can be done by right-clicking the latent. Edges represent virtual weights between latents, with blue indicating positive weights and orange indicating negative weights; edge thickness is proportional to weight magnitude. The toolbar provides functions for deleting nodes, saving/loading canvas layouts, exporting images, and toggling fullscreen mode.

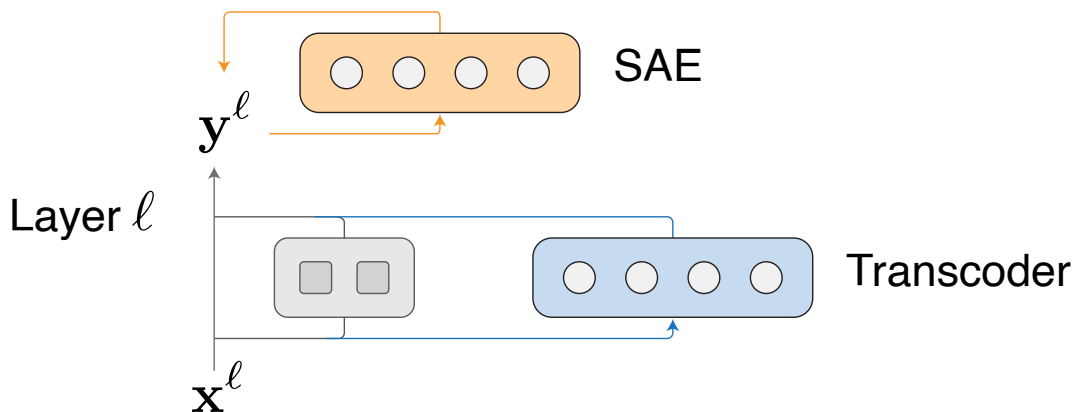


Figure 15. A simplified schematic of the differences between SAEs and transcoders. SAEs are designed to take in $\mathbf{y}^\ell$ and reconstruct $\hat{\mathbf{y}}^\ell$. Transcoders are designed to take in $\mathbf{x}^\ell$ and compute $\hat{\mathbf{y}}^\ell$.